Ekansh Simgekar

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EDUCATION

Morris County School of Technology, Academy for Computer and Information Sciences — Class of 2028

GPA: 4.1697 unweighted, 4.4329 weighted | **SAT:** 1560 (RW 770, Math 790) | **PSAT:** 1470 (RW 710, Math 760)

Coursework: AP Computer Science Principles (5, self-studied), AP Computer Science A (in progress), Computer Programming I & II (dual enrollment, Fairleigh Dickinson University), Calculus Honors, Physics Honors, Chemistry Honors, Pre-Calculus Honors, Computer Science Programming I Honors, Software Design, Introduction to Java Programming, Emerging Technologies (cybersecurity, AI)

EXPERIENCE

Robotics Team President (2026–present), **Mechanical Leader** (2025–26) — Sept 2024 – Present

FIRST Robotics Competition (FRC) and FIRST Tech Challenge (FTC), Morris County School of Technology

- Wired a swerve-drive FRC robot (four NEO / NEO 550 modules); team placed 8th at its first FIRST Mid-Atlantic event
- Received the Rising All-Star Award (2026)
- Teach new members the engineering design process, CAD, and wiring; about 10 hours per week in season

Technology Student Association (TSA) — Sept 2024 – Present

- 4th place in New Jersey, Technology Bowl (2025) and Geospatial Technology (2026)
- Wrote Python (rasterio, NumPy, Matplotlib) to map NYC urban heat; combined four public datasets and showed AC access alone explained little of the heat variation
- 2026–27 events: Data Analytics, Automated Manufacturing Prototype, Manufacturing Prototype

Student Advocacy Panel Member — Sept 2026 – Present

- Gather input from classmates and push for changes to school services

Student Council Secretary — Sept 2025 – June 2026

- Represented students, raised over $1,000 through homecoming and pep rallies, and attended leadership conferences

STEM Camp Counselor, MCST STEM Summer Camp and Essex County 4-H — Summer 2025

- Taught children Onshape and Arduino, ran a robotics coding station, and helped run engineering competitions (30 volunteer hours)

PROJECTS

FireWatch, Congressional App Challenge — 2026

- Wildfire monitor: Raspberry Pi cameras and an MLX90640 thermal sensor flag possible fires with OpenCV; a FastAPI server (deployed on Render) maps alerts beside NASA FIRMS satellite data and routes users around fires with OpenRouteService
- Designed owner confirmation for every alert to limit false alarms

Joystick-Controlled Prosthetic Hands, Hack Club Fallout — 2025 – 2026

- Built two versions: a cardboard servo prototype, then NEMA 17 stepper motors with adjustable fingers and wrist rotation (Arduino, C++, Onshape)
- Redesigned the string routing inside each finger to fix weak finger movement; certified for 60 hours of work

LunchLoop, Hack Club Capitol hackathon — 2026

- Joined a team of strangers to build a cafeteria demand forecaster (Python, pandas, FastAPI, React) using recent-day averages and a newsvendor safety buffer, tested on held-out weeks
- On simulated data, found $1,000+ in savings and about 47% less food waste; certified for 36 hours

Gamlect, coral gamete collector — 3 months

- Led CAD for a 3D-printed collector built with the Academy of Biotechnology, iterating on measurements through test prints

Plant Health Monitor, personal project

- Built an Arduino sensor system with an LED display that alerts when soil, water, light, or humidity is off; still in daily use at home

SERVICE

Mind4Youth, teen mental health organization — Oct – Nov 2024

- Designed and produced positive-affirmation cards distributed to the community (23 volunteer hours)

Yearbook Staff — 2026 – 2027

SKILLS

Languages: Java, Python, C++, JavaScript, HTML/CSS | **Hardware:** Arduino, Raspberry Pi, wiring, 3D printing | **Tools:** Onshape, TinkerCAD, FastAPI, pandas, OpenCV, Git/GitHub | **Spoken:** English, Spanish (Spanish III Honors)